

The Two Faces of Time

Coordinate Time, Proper Time, and Accumulated Decoherence

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<https://github.com/hmahaffeyges/IAM-Validation>

March 2026; revised October 2026

Zenodo DOI: 10.5281/zenodo.18702042

Abstract

Time enters physics in more than one role. The Machian relational framework [Barbour & Bertotti, 1982] showed that coordinate time is a redundant label on the spacetime manifold. We distinguish three quantities that are often treated interchangeably: (i) coordinate time, a label with no preferred direction and no accumulation; (ii) proper time τ , the geometric path length along a worldline, which does not by itself distinguish past from future; and (iii) accumulated decoherence, measured in the Informational Actualization Model (IAM) by the activation function $E(a) = \exp(1 - 1/a)$: thermodynamic, irreversible, and carried only by massive particles. Photons travel null geodesics, $d\tau = 0$: they accumulate neither proper time nor decoherence. Massive particles accumulate both and constitute the matter sector. The sector separation follows from the causal structure of the manifold combined with the thermodynamic cost of irreversible decoherence on timelike worldlines. Within this picture the arrow of time is set by the irreversibility of each decoherence event, each of which carries a minimum cost $k_B T \ln 2$ per bit [Landauer, 1961]; $E(a)$ is monotonically increasing. We show how the distinction bears on the Hubble tension and how it sits within shape dynamics.

Keywords: time; accumulated decoherence; Machian relationism; arrow of time; electroweak symmetry breaking; gravitational decoherence; sector separation; Informational Actualization Model

1 Two Faces

Time is two-faced.

Barbour identified the first face in 1982 with the Machian relational framework: coordinate time is redundant, a label on a dimension, not a river carrying us forward [Barbour & Bertotti, 1982]. That result has shaped how foundational questions about time are posed.

The second face concerns irreversibility. Is every temporal quantity equally redundant? We argue that one is not: the accumulated count of irreversible decoherence events on a worldline has a physical carrier and a direction, and the relational framework accommodates it naturally.

2 Three Quantities Worth Distinguishing

Time appears in physical theory in at least three distinct roles that are worth distinguishing carefully. The differences between them are not merely terminological.

2.1 Coordinate time: the map

Coordinate time, t or the scale factor a , is a label on the spacetime manifold. It assigns a number to each event. It does not require anything to happen. It does not require worldlines to exist, mass to be present, or irreversible processes to occur. The manifold exists with its coordinate structure whether or not any event within it is irreversible.

Barbour’s central insight [Barbour & Bertotti, 1982, Barbour, 1994] is that this coordinate has no privileged status. It is redundant as a fundamental concept. The relational framework constructs dynamics from configurations alone, without reference to an external temporal parameter. Coordinate time is the map. It is not the territory.

2.2 Proper time: path length on the map

Proper time τ is the geometric path length along a worldline:

$$\tau = \int \sqrt{-g_{\mu\nu} dx^\mu dx^\nu}. \quad (1)$$

It is still a geometric quantity, a property of the spacetime manifold measured along a particular path. A clock measures proper time. It tells you how far you have traveled on the map.

Critically, proper time is reversible in principle. In a universe with time-reversal symmetric laws, a worldline running forward and backward in coordinate time accumulates τ in both directions. Proper time distinguishes clocks from one another but does not, by itself, distinguish past from future.

For null geodesics, photons, $d\tau = 0$ identically. Photons plot on the manifold but accumulate zero proper time. This is not an approximation. It is the definition of a null

geodesic.

2.3 Accumulated decoherence: the record

Accumulated decoherence is different from both coordinate time and proper time. It is not a geometric quantity. It is thermodynamic.

It counts the irreversible quantum-to-classical transitions on a worldline, the decoherence events that write classical information on the cosmic horizon at a Landauer cost $k_B T \ln 2$ per bit [Landauer, 1961]. It requires:

1. massive particles on timelike worldlines ($d\tau > 0$);
2. irreversible quantum-to-classical transitions (gravitational decoherence).

Massive particles exist from electroweak symmetry breaking ($T \approx 100$ GeV, $t \approx 10^{-11}$ s), when the W , Z and the charged fermions acquire mass [Mahaffey, 2026c]. The irreversibility itself does not depend on CP violation: decoherence and entropy production are irreversible for any interaction. $E(a)$ is negligible for most of cosmic history ($E(z=10) = 4.5 \times 10^{-5}$) and grows as structure forms.

The geometric and thermodynamic descriptions are shown in Figure 1. The same worldline admits a geometric and a thermodynamic description.

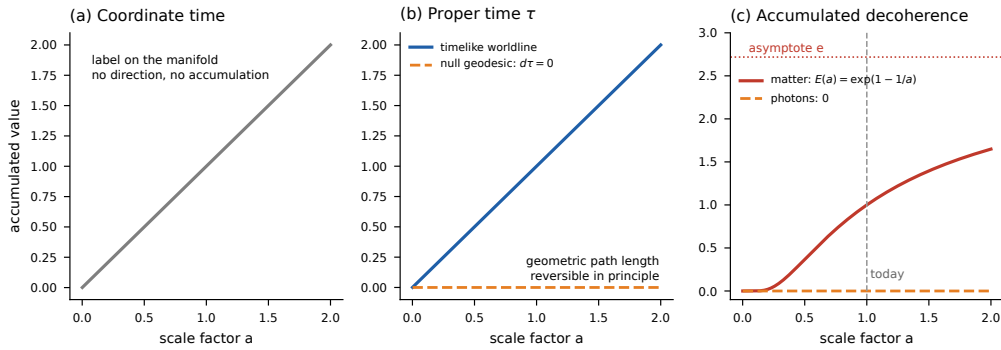


Figure 1: The two faces of time. Panel (a): geometric quantities, coordinate time (redundant label, Barbour & Bertotti 1982) and proper time τ (path length along worldline). Photons plot on the manifold but accumulate neither. Panel (b): accumulated decoherence, $E(a) = \exp(1 - 1/a)$, a thermodynamic quantity distinct from both: irreversible and carried by massive particles only. The matter sector and photon sector are separated by construction.

3 The Second Face: Accumulated Decoherence

3.1 The activation function

Within the Informational Actualization Model (IAM, Mahaffey 2026a), accumulated decoherence is measured by the activation function:

$$E(a) = \exp\left(1 - \frac{1}{a}\right), \quad (2)$$

where a is the cosmological scale factor normalized to unity today.

The properties of $E(a)$ follow directly from its physical interpretation:

- $E(a) \rightarrow 0$ as $a \rightarrow 0$: negligible in the early universe.
- $E(1) = 1$: normalized to today.
- $E(a) \rightarrow e$ as $a \rightarrow \infty$: a finite asymptote, approached but never attained.
- $dE/da > 0$ always: monotonically increasing, one direction only.

The monotonic increase is not imposed. It follows from the irreversibility of each decoherence event. Each bit written on the horizon cannot be unwritten. The record only accumulates. This is the arrow of time, not statistical, not initial-condition-dependent, but thermodynamically necessary.

3.2 Photons do not accumulate decoherence

Photons travel null geodesics. $d\tau = 0$ by construction. No proper time accumulates. No gravitational decoherence occurs; decoherence requires nonzero proper time on a timelike worldline. Therefore:

$$E(a)|_{\text{photon}} \equiv 0. \quad (3)$$

This is the physical basis for the sector split. Photons interact with the spacetime geometry; they are bent by gravity, redshifted by expansion, but they add nothing to $E(a)$. The modification to gravitational dynamics that $E(a)$ produces applies exclusively to the matter sector.

3.3 The Machian framework already knew this

The relational framework insists that physics be built from observable relations between things. Coordinate time fails this test, it is not directly observable. Proper time passes: clocks measure it. Accumulated decoherence passes as well, and it is the temporal quantity that distinguishes past from future through a physical mechanism rather than a statistical assumption.

Mach’s criterion, “time is an abstraction at which we arrive through the changes of things”, selects physical change over coordinate time. Not all changes are equal: reversible changes leave $E(a)$ unchanged; only irreversible decoherence events, each paying its Landauer cost, advance it.

4 What the Sector Split Implies

The distinction between the geometric and thermodynamic quantities has an immediate observational consequence that may help clarify a persistent tension in cosmological observations.

The CMB measurement of H_0 uses photons emitted at last scattering, traveling null geodesics for 13.8 billion years. The entire measurement is geometric. No proper time accumulated and no decoherence recorded. It reads the geometry. $H_0^{\text{photon}} = 67.16 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is what the manifold says about its own expansion rate [Planck Collaboration, 2020].

The local distance ladder uses Cepheids, Type Ia supernovae, and TRGB stars, massive objects on timelike worldlines that have accumulated decoherence for billions of years. It reads the matter sector. $H_0^{\text{matter}} = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is what the matter sector says about its own expansion rate [Riess et al., 2022, Mahaffey, 2026b].

Within this framework, these are not necessarily two measurements of the same quantity. They may be measurements of two distinct quantities, one geometric and one thermodynamic, that a purely geometric treatment does not distinguish. The Hubble tension is consistent with the sector split, and the predicted values are derived from first principles with zero free parameters beyond ΛCDM .

The predicted split follows from a single coupling constant derived from the virial theorem:

$$\beta_m = \frac{\Omega_m}{2} = 0.15765, \quad (4)$$

with zero free parameters beyond ΛCDM [Mahaffey, 2026a].

5 The Relational Framework and Shape Dynamics

Shape dynamics [Barbour, 2011, Barbour, Koslowski & Mercati, 2014] replaces four-dimensional spacetime refoliation invariance with three-dimensional conformal invariance. The universe is described by the evolution of shape, the relational configuration of matter, without reference to absolute scale or absolute time.

IAM is compatible with this framework at every level. IAM does not modify the background geometry. It does not alter the Friedmann equations. It operates at the perturbation level, adding a single Hubble friction term to the matter perturbation equa-

tions:

$$\ddot{\delta}_m + 2H\dot{\delta}_m - \frac{3}{2}\Omega_m H^2 \mu(a) \delta_m = 0, \quad (5)$$

where

$$\mu(a) = \frac{H_{\Lambda\text{CDM}}^2}{H_{\Lambda\text{CDM}}^2 + \beta_m E(a)} < 1. \quad (6)$$

The geometric arena (shape space, the relational configuration) is untouched. Accumulated decoherence is what happens on the worldlines *within* that arena; it does not modify the arena itself. Shape dynamics asks what the universe’s configuration is; IAM asks how much irreversible information the matter sector has produced in reaching it. The two questions are complementary.

6 An Open Question

Electroweak symmetry breaking selects one point of the degenerate Higgs vacuum manifold. Is that selection itself the first gravitational-scale decoherence event, with its own Landauer cost at the transition temperature? This remains an open question [Mahaffey, 2026c].

7 Repository

The complete thermodynamic derivation is developed in the IAM theory paper [Mahaffey, 2026a]. The full collection of papers addressing observational consequences, foundational extensions, open problems, and future predictions is available in the documents folder of the repository at <https://github.com/hmahaffeyges/IAM-Validation>. Follow whichever thread interests you.

Acknowledgments

The debt to Jacobson [1995], Cai and Kim [2005], and Barbour and Bertotti [1982] is explicit throughout. This work builds directly on their foundational contributions.

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